DeepFakes Detection





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BiDA Lab

Biometrics & Data Pattern Analytics Lab

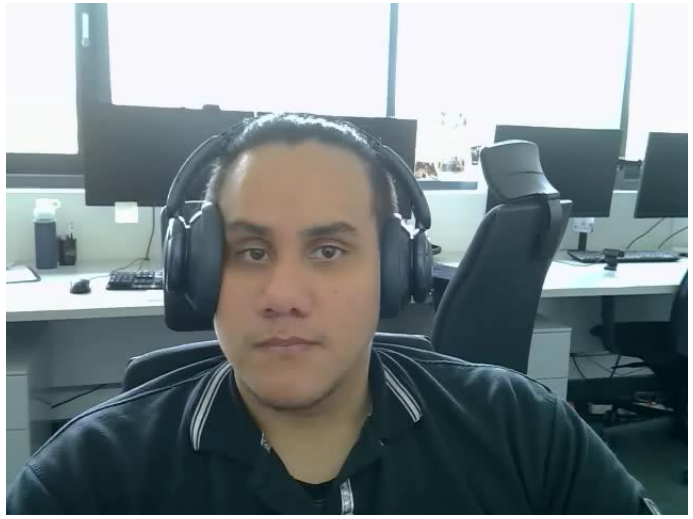
UAM

Universidad Autónoma de Madrid

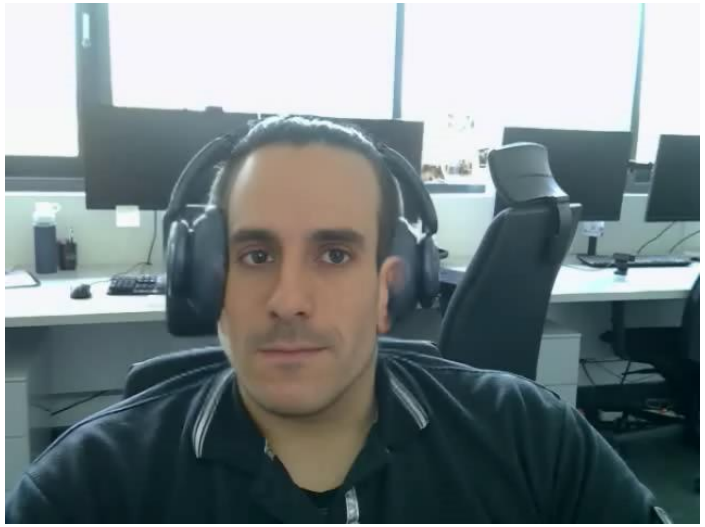
What is it?

Media Forensics has attracted a lot of attention in the last years in part due to the increasing concerns around DeepFakes.

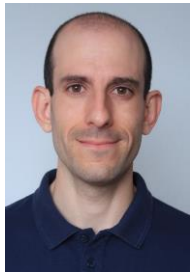
Real Video



DeepFake Video



Data Available:
Only 1 Picture

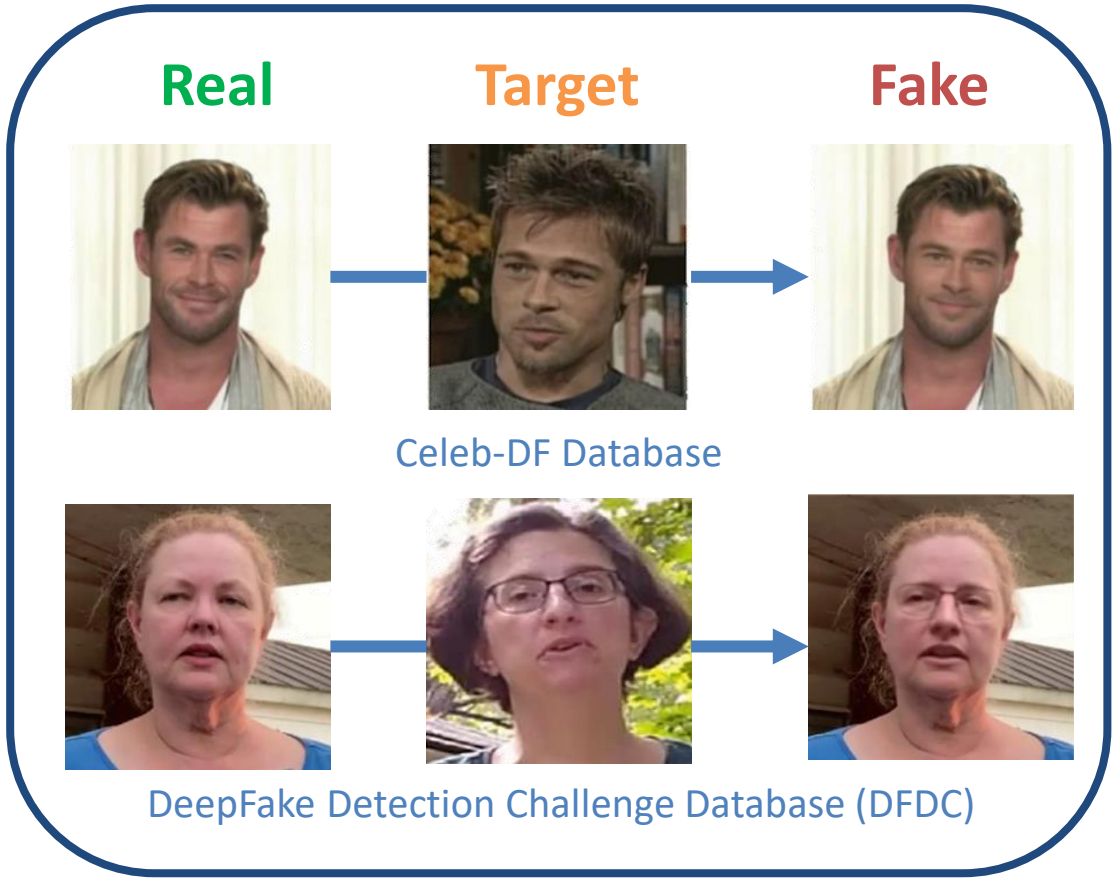


Target

- R. Tolosana, R. Vera-Rodriguez, J. Fierrez, A. Morales and J. Ortega-Garcia, "DeepFakes and Beyond: A Survey of Face Manipulation and Fake Detection," Information Fusion, vol. 64, pp. 131-148, 2020.

What is it?

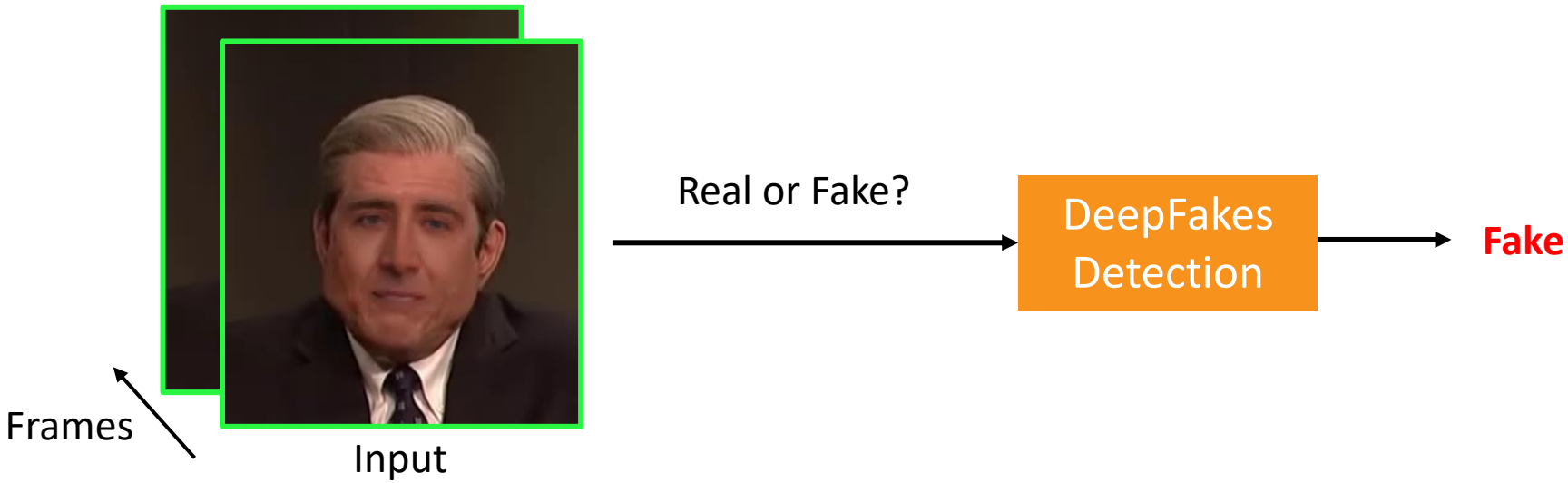
DeepFake (Identity Swap) is referred to a deep learning based technique able to create fake videos by swapping the face of a person by the face of another person.



- R. Tolosana, R. Vera-Rodriguez, J. Fierrez, A. Morales and J. Ortega-Garcia, "DeepFakes and Beyond: A Survey of Face Manipulation and Fake Detection," Information Fusion, vol. 64, pp. 131-148, 2020.

DeepFakes Detection

Detection of digital face manipulations.



- R. Tolosana, R. Vera-Rodriguez, J. Fierrez, A. Morales and J. Ortega-Garcia, "DeepFakes and Beyond: A Survey of Face Manipulation and Fake Detection," Information Fusion, vol. 64, pp. 131-148, 2020.

TASK 1

DeepFakes Detection: Intra-Database Analysis



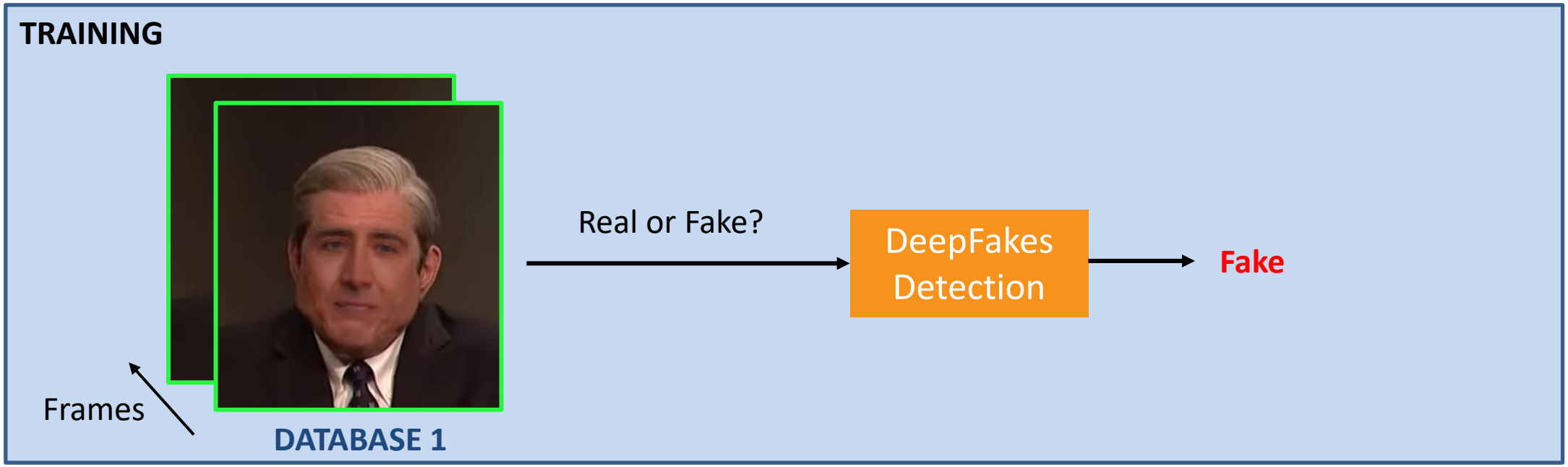
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Task 1 - DeepFakes Detection: Intra-Database Analysis

Intra-database analysis: we are going to train and evaluate the systems using the same database.

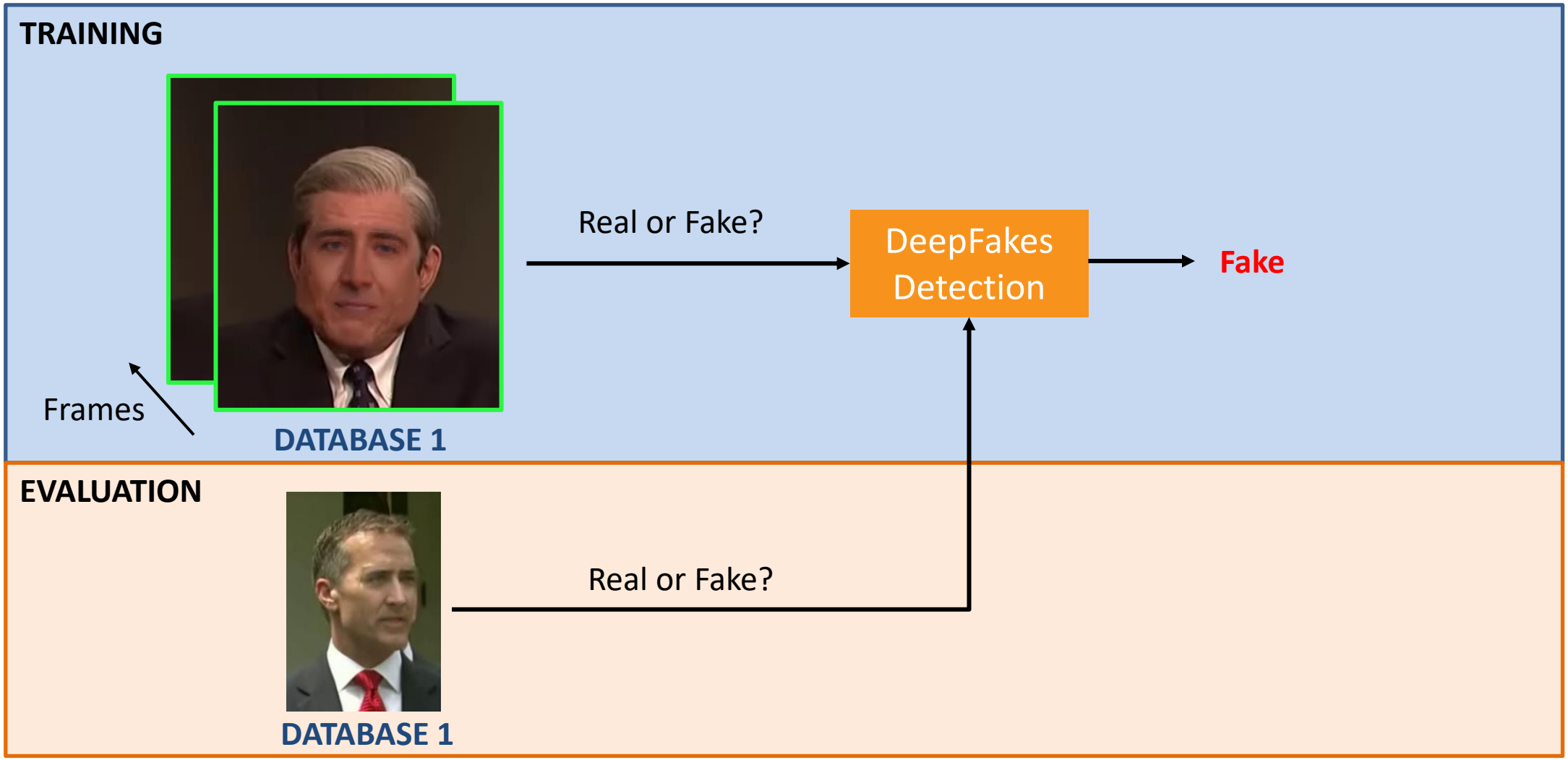
Task 1 - DeepFakes Detection: Intra-Database Analysis

Intra-database analysis: we are going to train and evaluate the systems using the same database.



Task 1 - DeepFakes Detection: Intra-Database Analysis

Intra-database analysis: we are going to train and evaluate the systems using the same database.



Task 1 - DeepFakes Detection: Intra-Database Analysis

UADFV database: 49 real and fake videos (each) using the FakeAPP software application (Computer Graphics).



- Y. Li, M. Chang, and S. Lyu, “In Ictu Oculi: Exposing AI Generated Fake Face Videos by Detecting Eye Blinking,” in Proc. IEEE Int. Workshop on Information Forensics and Security, 2018.

Task 1 - DeepFakes Detection: Intra-Database Analysis

Experimental protocol:

- Development: this dataset is considered only for the training of the systems.
  - Real folder: 38 videos.
  - Fake folder: 38 videos.

- Evaluation: this dataset is only considered for the final evaluation of the system (not for training!!!).
  - Real folder: 11 videos.
  - Fake folder: 10 videos.

For each video, we provide 10 frames.

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Task 1 - DeepFakes Detection: Intra-Database Analysis

The task is to classify each image as real or fake independently, NOT VIDEOS. Therefore, for each image you will have to provide a classification score and a decision.



Input Image

Real or Fake?

DeepFakes Detection

Score: 0.93 (Real)



Input Image

Real or Fake?

DeepFakes Detection

Score: 0.27 (Fake)

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TASK 2

DeepFakes Detection: Inter-Database Analysis



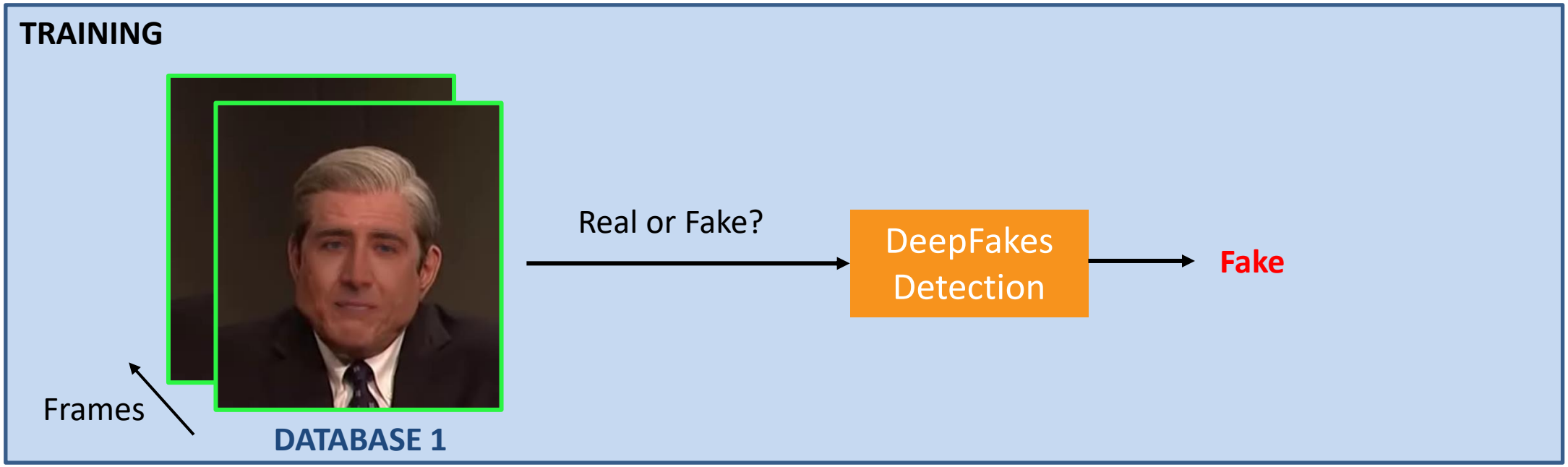
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Task 2 - DeepFakes Detection: Inter-Database Analysis

Inter-database analysis: we are going to train and evaluate the systems using different databases.

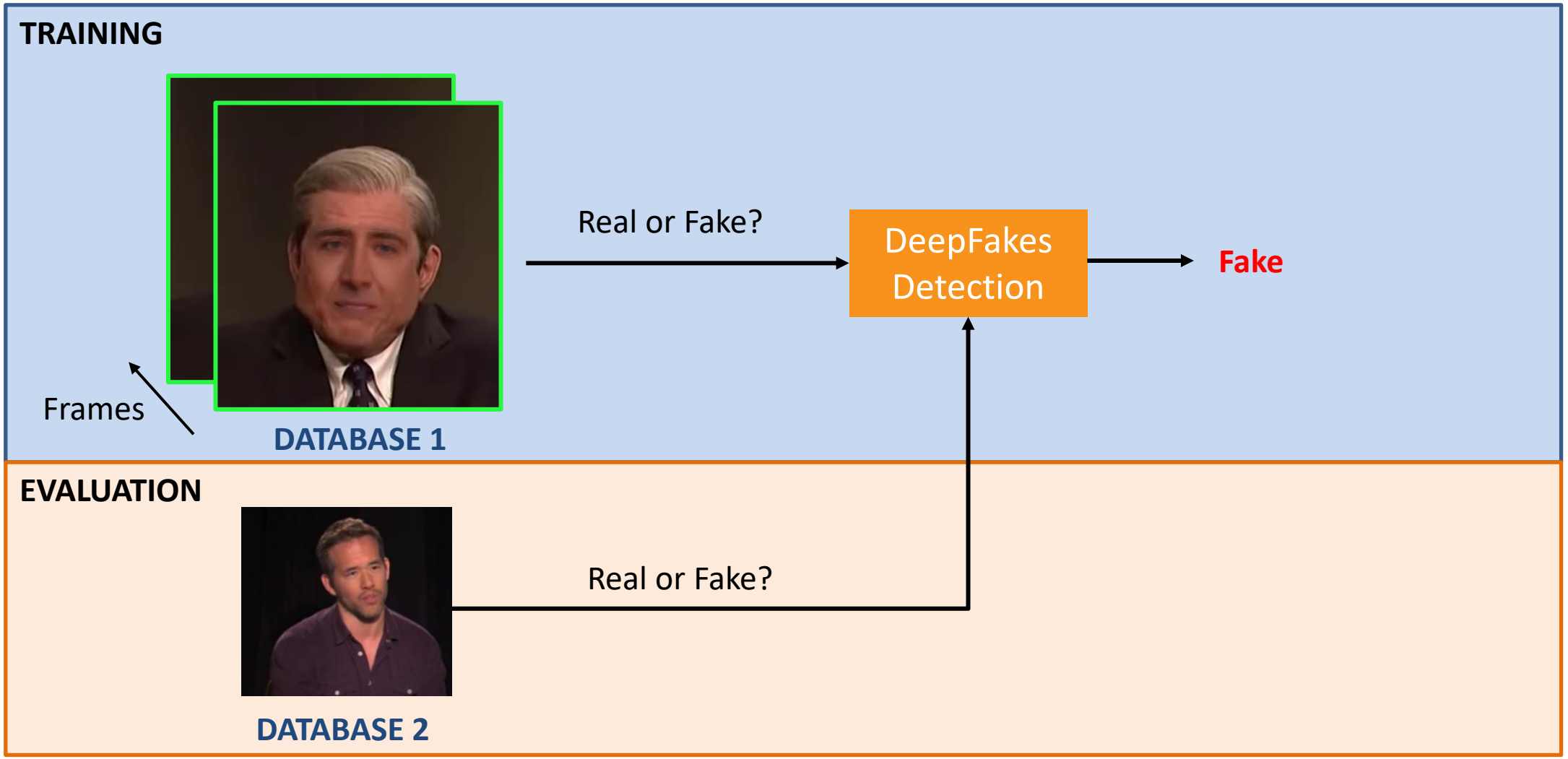
Task 2 - DeepFakes Detection: Inter-Database Analysis

Inter-database analysis: we are going to train and evaluate the systems using different databases.



Task 2 - DeepFakes Detection: Inter-Database Analysis

Inter-database analysis: we are going to train and evaluate the systems using different databases.



Task 2 - DeepFakes Detection: Inter-Database Analysis

Celeb-DF database (v1): 408 real and 795 fake videos using Deep Learning (Autoencoders).





- Y. Li, X. Yang, P. Sun, H. Qi, and S. Lyu, “Celeb-DF: A LargeScale Challenging Dataset for DeepFake Forensics,” in Proc. IEEE/CVF Conf. on Computer Vision and Pattern Recognition, 2020.

Task 2 - DeepFakes Detection: Inter-Database Analysis

Experimental protocol:

- Development: the same detection system developed in Task 1 (using only the UADFV database).

- Evaluation: only the Celeb-DF database is considered for the final evaluation of the system (not for training!!!).

  - Evaluation folder:

    - Real folder: 30 videos.

    - Fake folder: 30 videos.

For each video, we provide 10 frames.

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TASK 1

DeepFakes Detection:
Intra-Database Analysis

TASK 2

DeepFakes Detection:
Inter-Database Analysis



Restrictions: It is forbidden to use **pre-trained models**.
Only the **development dataset** can be used.

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TASK 3

DeepFakes Detection: Inter-Database Proposal



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Task 3 - DeepFakes Detection: Inter-Database Analysis

Goal: propose a new DeepFake detection system in order to achieve better inter-database results.

Task 3 - DeepFakes Detection: Inter-Database Analysis

Goal: propose a new DeepFake detection system in order to achieve better inter-database results.

- Development: no restrictions.

- Evaluation: only the Celeb-DF database is considered for the final evaluation of the system (not for training!!!).
  - Evaluation folder:
    - Real folder: 30 videos.
    - Fake folder: 30 videos.

For each video, we provide 10 frames.

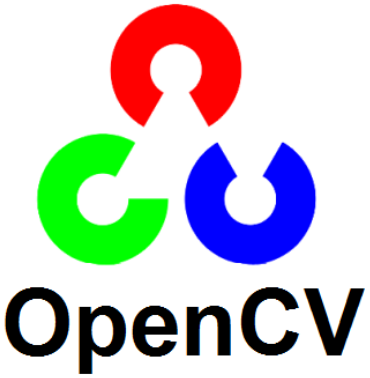
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Tips

Software and databases: no restrictions.













Evaluation

This Lab is carried out in groups of 4 people.

Evaluation of the Lab:

- Report (max. 8 pages): complete the Word template provided including the information and results of your proposed DeepFake detection system. Do not forget to include all your code when submitting via Moodle.

- Oral presentation (max. 10 minutes): each group should explain the proposal and results achieved in each of the three Tasks.

- The report and presentation must be sent via Moodle the day before the oral presentations in class (check specific deadline in Moodle).

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